

Homework #6 (110 points + 30 bonus points)

207.547 Seminar in Experimental Psychology: Computational Modeling
Spring 2026

Due: Tuesday, May 26th, at 11:59 pm

1. (30 pts). Simulation and parameter recovery using a simple reinforcement learning model

- (1) (10 pts) Find the `simulate_hw6_model1.R` file and generate simulated data (`simul_data_hw6_model1.txt`) using the code and its default settings (`num_subjs = 30, num_trials = 300`).
- (2) (20 pts) Using `hw6_model1.stan` and `hw6_model1.R` files,
 - (2a) (5 pts) Find the posterior distribution of individual and group (hyper) parameters.
 - (2b) (15 pts) Draw a scatterplot of true parameters and estimated parameters (x axis: true individual parameters, y axis: individual posterior means). Add error bars along the y-axis (error bars = ± 1 SD of individual posterior distributions). Check if true parameters are relatively well recovered.

2. (20 pts) Comparing the outputs from hierarchical and non-hierarchical Bayesian models.

- (1) (5 pts) Program your R and Stan codes that will estimate individual posterior distributions of alpha and beta (a non-hierarchical version). Use the same simulated data generated in Question#1.
- (2) (10 pts) Draw a scatterplot of true parameters and estimated parameters (x axis: true individual parameters, y axis: individual posterior means). Add error bars along the y-axis (error bars = ± 1 SD of individual posterior distributions).
- (3) (5 pts) Do you see a noticeable difference in the two scatter plots you generated in Question#1(2) and Question#2(2)? Discuss why.

3. (30 pts) “Shrinkage” in hierarchical Bayesian data analysis

- (1) (10 pts) In the `simulate_hw6_model1.R` file, set `num_subjs = 200, num_trials = 100` and generate 200 fake subjects’ data.
- (2) (10 pts) Find the posterior distribution of individual and group (hyper) parameters (**warning: MCMC sampling may take a long time**).
- (3) (10 pts) Draw a scatterplot of true parameters and estimated parameters (x axis: true individual parameters, y axis: individual posterior means). Add error bars along the y-axis (error bars = ± 1 SD of individual posterior distributions). Check if true parameters are relatively well recovered.
- (4) (Bonus point: 10 pts) Do you notice “shrinkage” in one parameter but not so much in the other parameter? Can you explain why?

4. (30 pts) A model with two learning rates

Modify provided Stan and R codes so that we can allow two learning rates in the reinforcement learning model. `alpha_pos` $\rightarrow$ learning rate when a prediction error is zero or above. `alpha_neg` $\rightarrow$ learning rate when a prediction error is less than zero. Use the following settings for generating true parameters:

```
simul_pars = data.frame(alpha_pos = rnorm(num_subjs, 0.20, 0.08),
                        alpha_neg = rnorm(num_subjs, 0.30, 0.10),
                        beta = rnorm(num_subjs, 2.00, 0.70),
                        subjID = 1:num_subjs)
```

Also, set `num_subjs = 30`, `num_trials=100`.

- (1) **(5 pts)** Find the posterior distribution of individual and group (hyper) parameters.
- (2) **(5 pts)** Draw a scatterplot of true parameters and estimated parameters (x axis: true individual parameters, y axis: individual posterior means). Add error bars along the y-axis (error bars = ± 1 SD of individual posterior distributions). Check if true parameters are relatively well recovered.
- (3) **(10 pts)** In the `simulate_hw6_model1.R`, change `num_trials`, `pr_correct_option1`, and `pr_correct_option2` (lines 9-11) like the following and re-generate simulated data:

```
num_trials <- 300
pr_correct_option1 <- 0.3
pr_correct_option2 <- 0.6
```

Find the posterior distributions and draw a scatter of true parameters and estimated parameters (x axis: true individual parameters, y axis: individual posterior means). Add error bars along the y-axis (error bars = ± 1 SD of individual posterior distributions).

- (4) **(10 pts)** Do you notice any difference between what you found in (2) and (3)? Try to explain what made the difference.

5. (Optional 30 pts) In HW#2, you estimated model parameters of the *ra_prospect* model using MLE. In this homework, use grid approximation to approximate posterior distributions of the parameters.

- (1) **(20 pts)** Based on lecture materials and chapters from John Kruschke's book, use grid approximation and calculate posterior distributions of the *ra_prospect model*. Use 10 grids per parameter. Plot and attach your posterior distributions. Attach your R (or Python/Matlab) code.
- (2) **(10 pts)** Use 30 grids per parameter. Plot and attach your posterior distributions. Attach your R (or Python/Matlab) code.

Attach all your R/Stan program. Please generate a single PDF file with all the outputs.